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Experimental analysis on the rapid measurement of a high precision Brillouin scattering spectrum in water using a Fabry–Pérot etalon

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Abstract

The Brillouin lidar system with a Fabry–Pérot (F–P) etalon has been widely used in remote sensing to study ocean characteristics such as temperature, sound speed, etc. In this system, the measurement error stemming from the spectral broadening of an F–P etalon was typically neglected in applications where high measurement accuracy was not essential. In this paper, an integration method on radius and a convolution method on frequency are proposed to remove the broadening effect and improve the measurement accuracy, respectively. Experiments of underwater Brillouin scattering show that both fitting methods effectively reduce the measurement error due to the transmission function of an F–P etalon. A temperature accuracy of 0.1 °C and a salinity accuracy of 0.5‰ were achieved. Moreover, with the convolution method, the computational cost was reduced to 3 s, which enables rapid measurement of Brillouin shift and linewidth and makes the on-line remote sensing applications possible.

Keywords: Brillouin scattering, lidar, Fabry–Pérot, spectrum measurement

(Some figures may appear in colour only in the online journal)

1. Introduction

Brillouin scattering has been demonstrated as a versatile tool in studying transparent media, such as optical fibers [1–3], air [4–8] and water [9–11]. The spectrum accuracy obtained from Brillouin scattering is important since the spectrum characteristics are related to many physical properties of the studied media. The temperature and stress in the optical fibers were retrieved by measuring the shift and the energy of the Brillouin scattering [12], and the Brillouin shift and linewidth were used to measure the ocean temperature, salinity and viscosities, etc in ocean remote sensing applications [9, 13, 14].

Some spectrum detection techniques were used to obtain the Brillouin spectral characteristics. In current methods, the

edge technology monitors the energy variation instead of the full spectrum [15]. The scanning Fabry–Pérot (F–P) interferometers [9] work well in the laboratory and the frequency distribution of the Brillouin scattering light can be obtained by changing the plate spacing of the F–P interferometers to scan the whole spectrum step-by-step. This method fails to work in practical remote sensing applications because of the complexity of the precise controlling of the plate spacing and the expensive time cost on the scanning F–P interferometer. To overcome the aforementioned two problems, the F–P etalon, with a fixed plate spacing and without the need of scanning the spectrum step-by-step to obtain the full spectrum, was recently introduced to receive the Brillouin scattering echo signal [14]. Its capability of receiving the echo signal in real time makes it an active research field lately [14, 16].

However, instead of the Brillouin scattering spectrum that is normally used for measurement, the F-P etalon provides a 2D interference fringe pattern. This pattern consists of concentric rings, and the intensities on the radius direction form the interference spectrum. The Brillouin spectrum can be extracted through certain conversion procedures based on the interference fringe pattern. Previous works in this conversion treat the interference spectrum as the Brillouin scattering spectrum, and directly use the intensities on the radius direction for measurement. This method can be easily implemented in a short time. Unfortunately, the measurement accuracy of the characteristics is inevitably limited, as it neglects the effects due to the resolution limitation and the instrumental broadening of the F-P etalon. In addition, the effect of the noise in practice is also neglected from this method. The above effects enlarge the error from tens of MHzs to more than 100 MHz [17]. In ocean temperature measurement, for instance, this error of the temperature measurement accuracy is enlarged from about 0.5 °C to more than 1 °C [18, 19]. In some applications where high measurement accuracy is required, such as MHz magnitude or 0.2 °C, a higher measurement accuracy should be achieved without the loss of the capability of on-line measurement.

To achieve this, three problems should be addressed: the limitation due to the spectrum resolution, the error due to the noise and the effect due to the instrument broadening of the F-P etalon. Since our previous work [17] has effectively improved the spectrum resolution from more than 100 MHz–1 MHz magnitude and an ideal low-pass filter with little time-consumption can be used for denoising the spectrum, only the third problem needs to be addressed. Therefore, we propose some efficient methods to remove the broadening effect and improve the measurement accuracy in the short time level in this paper. With the proposed methods, the measurement with the broadening effect can be performed efficiently and the F-P etalon method can be implemented for the on-line measurement of physical properties of the water, which is very significant for the ocean remote sensing applications.

2. Theoretical analysis

The Brillouin scattering light of seawater is produced by the interaction between the incident light and the density variation of the water media. When this scattering light is collected and enters the F-P etalon, the interference phenomenon occurs.

This procedure is shown in figure 1 [20]. The Brillouin scattering light incidents the F-P etalon and is split into multiple beams by the two reflecting surfaces of the etalon. The reflecting procedure will produce an optical path difference between these beams. Therefore, when these beams are focused by the fringe forming lens, the interference phenomenon occurs on the image plane and the beams finally form the 2D interference fringe pattern, as shown in figure 2(a).

Figure 2(a) contains three orders of complete fringes. Each order of the fringe has three rings: the bright one in the middle is the Rayleigh ring, and the two less bright rings symmetrically located at both sides of the Rayleigh ring are

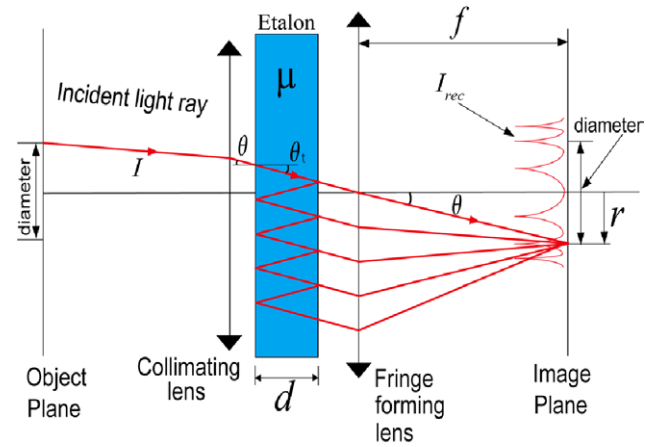


Figure 1. Schematic of the interference procedure with the F-P etalon.

the interference rings of the Stokes Brillouin scattering (the outer one) and the anti-Stokes scattering (the inner one). The interference spectrum is located on the radius direction. By extracting the intensities on the radius direction, the interference spectrum I_{rec} is shown in figure 2(b). The interference spectrum obtained from the F-P etalon is related to the radius r between the point and the fringe center, rather than the frequency. To mathematically describe the interference spectrum, a simple method is to set r as the variable, and calculate the intensity of the point with the distance r from the fringe center, i.e. $I_{rec}(r)$.

$I_{rec}(r)$ is the function of the scattering spectrum and the transmission function of the F-P etalon. Ideally the transmission function of the F-P etalon is an Airy function:

$$\begin{aligned} \text{Airy}(v, r) &= \frac{1}{1 + \frac{4F_R^2}{\pi^2} \cdot \sin^2 \left[\frac{2\pi\mu d v}{c} \cos \theta_t(r) \right]} \\ &= \frac{1}{1 + \frac{4F_R^2}{\pi^2} \cdot \sin^2 \left[\frac{2\pi\mu d v}{c} \sqrt{1 - \left(\frac{1}{\mu} \right)^2 \frac{r^2}{f^2 + r^2}} \right]} \end{aligned} \quad (1)$$

where v is the frequency of the incident scattering light, μ and d are respectively the refractive index and the plate spacing of the F-P etalon, c is the speed of light in the vacuum, θ_t is the refraction angle of the light in the etalon, f is the focus length of the lens, which is used to focus the interference light on the image plane. $F_R = \pi R^{1/2}/(1 - R)$ is the finesse, where R is the surface reflectance. The second term of equation (1) is similar to the transmission function of the scanning F-P interferometer. The F-P interferometer scans the interference spectrum and selects v through changing d . When the optical path difference $2\mu d \cdot \cos \theta_t$ is the integer multiple of the wavelength λ , i.e. $2\mu d \cdot \cos \theta_t = m\lambda = mc/v$, the interference light reaches the maximum intensity. So in the scanning interferometer, d is proportional to v , different d can select different v , and the second term of equation (1) is enough for measurement. However, for the F-P etalon, d is fixed, and the etalon selects different v through changing the refraction angle θ_t , i.e. v corresponds to θ_t . While θ_t is related to the

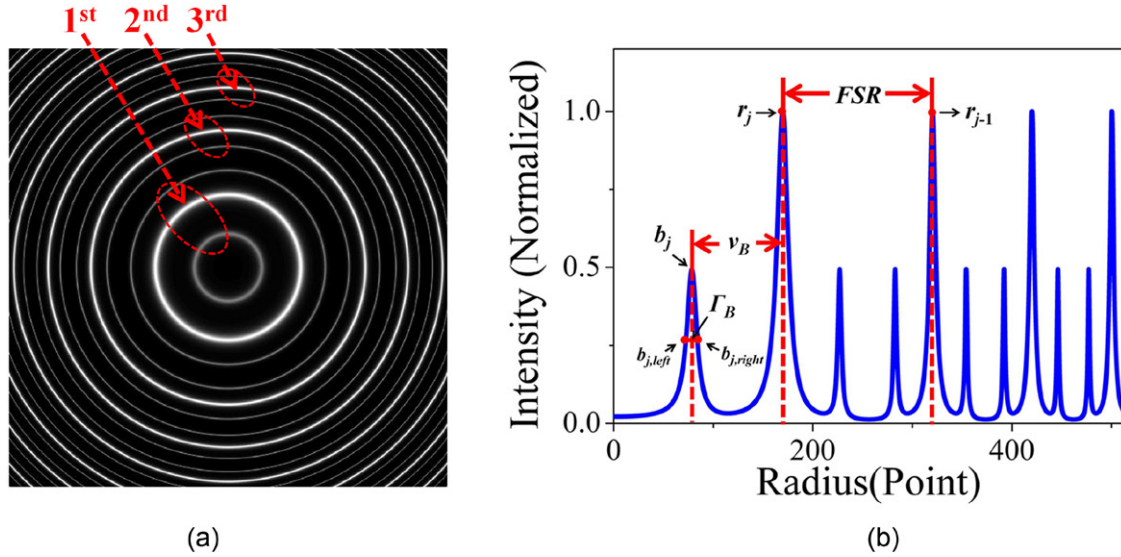


Figure 2. (a) Schematic of 2D interference fringe pattern. (b) 1D interference spectrum extracted from (a).

radius r , so different v is displayed at different radius r , and the Airy function for the F-P etalon can be expressed as the third term of equation (1).

When incident scattering light with various frequencies transmits through the F-P etalon, the interference intensity $I_{rec}(r)$ should be the sum of all intensities contributed by each single frequency, and can be expressed as an integration of the scattering spectrum $I(v)$ and the transmission function Airy(v, r) [21]:

$$I_{rec}(r) = \int_{-\infty}^{+\infty} I(v) \cdot \text{Airy}(v, r) dv \quad (2)$$

For Brillouin scattering, $I(v)$ consists of two peaks which are manifested as Lorentzian functions [15]:

$$I(v) = \frac{I_0 \cdot \Gamma_B}{\Gamma_B^2 + 4 \cdot (v - v_B)^2} + \frac{I_0 \cdot \Gamma_B}{\Gamma_B^2 + 4 \cdot (v + v_B)^2} \quad (3)$$

where, I_0 is constant that represents the maximum of the Brillouin peaks, v_B is the frequency shift of the Brillouin peak center from the center of the Rayleigh peak, named the Brillouin shift. Γ_B is the full width at half maximum (FWHM) of each Brillouin peak, named the Brillouin linewidth. The first term represents the Stokes Brillouin scattering, while the second represents the anti-Stokes Brillouin scattering.

In equation (2), the radius is related to the frequency. The relation is proved as follows [14, 22]:

$$\omega = \frac{r^2 - r_j^2}{r_j^2 - r_{j-1}^2} \cdot \text{FSR} \quad (4)$$

where ω is relative frequency, $\text{FSR} = c/(2\mu d)$ is the free spectral range of the F-P etalon, r_j and r_{j-1} is the center position of two neighboring Rayleigh peaks, as shown in figure 2(b). This relation is based on the fact that θ_i is small and can be sufficient for the approximation $\tan\theta_i \approx \sin\theta_i$. In fact, the focus length f is generally in meter magnitude, and is much larger than the radius r in μm magnitude, so the θ_i is small enough for the approximation. Therefore, previous works have already used

equation (4) to calculate the Brillouin shift and Brillouin linewidth. In figure 2(b), the Brillouin shift is represented by the distance between the Brillouin peaks and the Rayleigh peak in the same order, and the Brillouin linewidth is the FWHM of the Brillouin peak. By measuring the Brillouin peak position b_j and the half-maximum positions $b_{j,left}$ and $b_{j,right}$, the Brillouin shift and linewidth can be simply calculated through equation (4).

However, the scattering spectrum I is different from the interference spectrum I_{rec} , as the profile of I_{rec} has been changed by the transmission function Airy(v, r) of the etalon. The method with equation (4) neglects the Airy(v, r) and introduces a systematical error to the measurement, especially for the Brillouin linewidth measurement. To reduce this error, the effect of Airy(v, r) should be considered in the measurement, such as equation (1). Therefore, using equation (1) to fit the interference spectrum is a possible way, as this fitting has considered the Airy(v, r) and can measure a more accurate Brillouin shift and linewidth.

Equation (1) is an integration of scattering spectrum $I(v)$ and Airy(v, r) which has a complex form. For the fitting procedure, the time costs are mainly due to the numerical implementation on the integration and the complex calculation procedure on Airy(v, r). For instance, considering an M points sampling on v and an N points sampling on r , one step of fitting is implemented by $M \times N$ times calculation on integration and Airy(v, r) and M times on $I(v)$. Further, the calculation of Airy(v, r) costs more time than the integration, as the integration just consists of one multiplication and one addition, but Airy(v, r) contains more complex calculations than the integration, such as the sine function and others. The sine function is based on the Taylor expansion, and the approaching precision directly determines the orders that are needed. So, when facing the high precision requirement and high resolution spectra, the calculation of Airy(v, r) accounts for the major time costs.

To reduce this time cost, a possible way is to reduce the calculating times of the Airy function. In fact, the interference

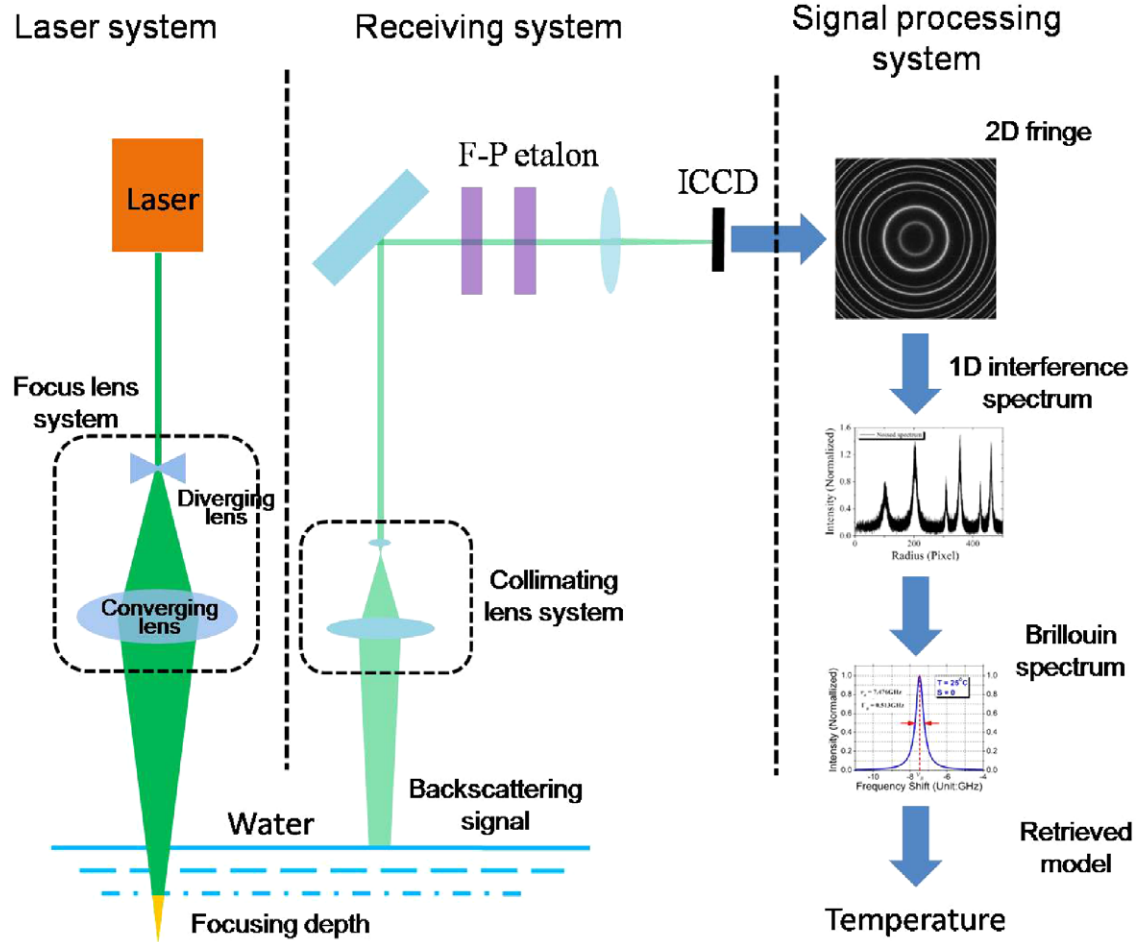


Figure 3. Schematic diagram of a Brillouin lidar system.

Table 1. Measured Brillouin shift and linewidth and the temperatures retrieved by the traditional methods.

ν_B (GHz)	Γ_B (GHz)	T_V (°C)	T_T (°C)	ΔT (°C)	S (%)
7.232	1.272	8.39	4.50	3.89	10.31

spectrum obtained by the F-P etalon is still related to frequency. The spectrum by radius is actually a special representation for the F-P interference. Therefore, once the spectrum by radius can be transformed to a normal spectrum by frequency, the calculations on the Airy function will only consider the frequency, and therefore will be reduced to M times, which will save time considerably.

For the F-P etalon, the optical path difference can be expressed as an integer times of the wavelength and a residual of less than λ , as follows:

$$2\mu d \cdot \cos \theta_i(r) = m \cdot \lambda + \Delta \cdot \lambda = 2\mu d + \Delta \cdot \frac{c}{\nu} \quad (5)$$

where m is the integer times, Δ is from -0.5 to 0.5 , which is independent from ν . $2\mu d = m\lambda$ is the occurrence conditions for the m th orders of interference intensity reaching maximum. The $\cos \theta_i(r)$ can be expressed as:

$$\cos \theta_i(r) = 1 + \frac{c}{2\mu d \nu} \cdot \Delta = 1 + \frac{\text{FSR} \cdot \Delta}{\nu} = 1 + \frac{\omega}{\nu} \quad (6)$$

where $\omega = 2\mu d \cdot \Delta / c = \text{FSR} \cdot \Delta$ is related to r , and the approximation of this relation is equation (4). By substituting equation (6) into equation (1), I_{rec} can be expressed as:

$$\begin{aligned} I_{\text{rec}}(\nu) &= \int_{-\infty}^{+\infty} I(\omega) \cdot \frac{1}{1 + \frac{4F_R^2}{\pi^2} \cdot \sin^2 \left[\frac{2\pi\mu d \nu}{c} \left(1 + \frac{\omega}{\nu} \right) \right]} d\omega \\ &= \int_{-\infty}^{+\infty} I(\omega) \cdot \frac{1}{1 + \frac{4F_R^2}{\pi^2} \cdot \sin^2 \left[\frac{2\pi\mu d}{c} (\nu + \omega) \right]} d\omega \\ &= I(\nu) \otimes \text{Airy}(\nu) \end{aligned} \quad (7)$$

As ω is from $-0.5 \cdot \text{FSR}$ to $0.5 \cdot \text{FSR}$ and the Airy function is symmetrical, $\nu + \omega$ can also be expressed as $\nu - \omega$. Then the I_{rec} is transformed as the convolution of $I(\nu)$ and $\text{Airy}(\nu)$. Therefore, we can first transform $I_{\text{rec}}(r)$ to $I_{\text{rec}}(\nu)$ with equation (4), then fit $I_{\text{rec}}(\nu)$ with equation (7). Obviously, this fitting only needs to calculate the Airy function with frequency ν , and it means the calculating times are reduced to M . Compared with the fitting procedure with equation (1), this fitting has more advantage on the time costs, while it only needs to add the procedure of transforming $I_{\text{rec}}(r)$ to $I_{\text{rec}}(\nu)$ with equation (4), which shows few time costs.

In the next section, under water experimental data will be carried out to verify the two fitting procedures. Then measurement accuracy of the Brillouin shift and linewidth are

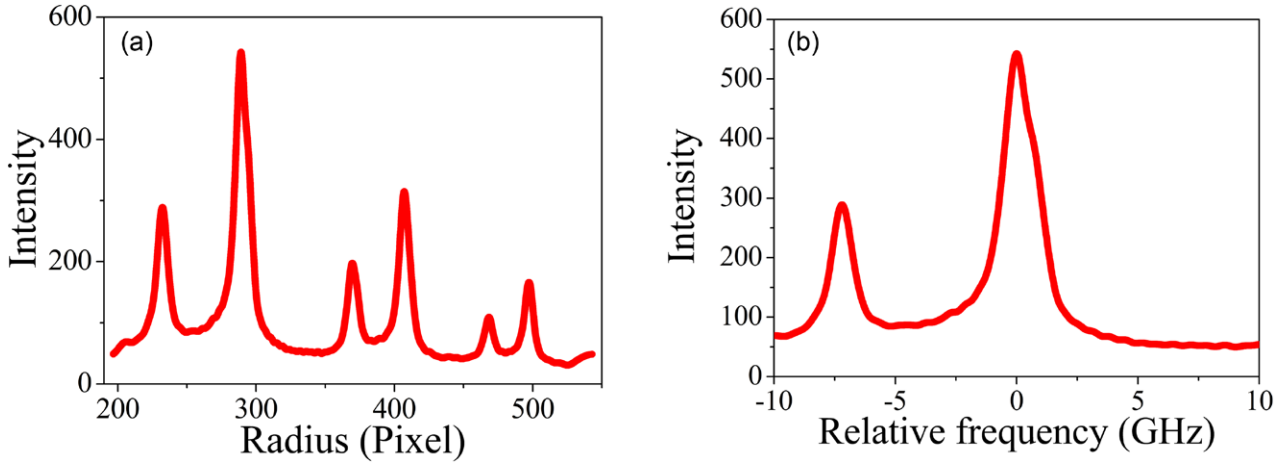


Figure 4. (a) The interference spectrum on radius; (b) the interference spectrum on frequency.

discussed. Based on the analysis, the real-time capability of the two fitting procedures is finally evaluated.

3. Experimental data

Measurements were carried out in a horizontal water tank in a laboratory. The experimental data is obtained in underwater experiments with a Brillouin lidar system using an F-P etalon and an intensified charge coupled device (ICCD), designed for measuring the Brillouin scattering signal of water at 532 nm [17, 20, 22]. Figure 3 shows the schematic diagram of a Brillouin lidar system. The laser beam passes through a focusing lens system and then is focused at a depth under water where the generated scattering signal can be collected by the telescope. Then, the F-P etalon receives this signal and transforms it into a 2D interference fringe pattern which is recorded by ICCD. A lens system consisting of a small diverging lens and a large converging lens was employed to determine the detected depth. The diverging lens was fixed on a plate driven by a DC electric motor. So the distance between the two lenses can be adjusted and the backscattering signal at a different depth can be obtained. It needs to be explained that this technique with a focusing lens system is great for a fixed distance, but that it would be very cumbersome for profiling measurements. Our next work is to improve the focusing lens system so that it can be applied in the real application.

The laser used in this system is a stable injection seed pulsed laser with a pulse energy of 650 mJ, and a repetition rate of 10 Hz. The F-P etalon is a CVI solid etalon made of quartz with a free spectral range of $FSR = 20$ GHz and a reflective rate of 99.5%. After calibration [20], the FSR was calibrated to 19.6 GHz. The fringe forming lens has a focus of around 0.5 m. The ICCD system consists of an ICCD camera and an ICCD controller. The ICCD camera is a Princeton Instruments PI-MAX III with a resolution of 1024×1024 and a pixel size of $12.8 \times 12.8 \mu\text{m}$. The ICCD controller is a Princeton Instruments ST-133.

It should be noted that there is a limited solid angle of acceptance in the high resolution F-P etalon [23]. There will be a maximum angle θ_m for which a ray will be transmitted within the F-P pass band:

$$\theta_m < \sqrt{\lambda/F_R d}$$

In this experiment, the value of $d = 0.5$ cm, $F_R = 627$ and $\lambda = 532$ nm, $\theta_m \approx 0.0235^\circ$. The diameter of the F-P etalon is 25.4 mm, and the corresponding depth is 62 m (2 m in air and 60 m in water). In order to more efficiently receive the return photons, a collimating lens system is used to make sure the backscattering light signal is incident within the acceptance solid angle for the F-P etalon.

The scattering signal is produced by stimulated Brillouin scattering. Unlike the spontaneous Brillouin scattering shown in the simulation in figure 2, the stimulated Brillouin scattering only contains two rings in each order: the inner one represents the Stokes Brillouin peak and the outer one represents the Rayleigh peak. The backscattered signal is collected at 180° with the optical system and then transmitted through the F-P etalon; the 2D interference fringes are recorded by ICCD.

The 2D fringes will be transformed to a 1D interference spectrum by using a data fusion strategy. Subsequently, the precise Brillouin spectrum can be extracted by a series of signal processing methods, and the Brillouin shift and linewidth can be obtained. Finally, temperature and salinity can be measured based on the theoretical retrieved model.

4. Analysis

The experimental data is first measured by traditional methods [14, 22]. Here the interference spectrum is extracted from the radius direction, and the measurement is held based on the relations in equation (4). Water temperature can be retrieved by measuring the Brillouin shift [18]. Recent studies show that there is a strong dependence of Brillouin linewidth on temperature rather than the shift [24]. The dependence of the Brillouin linewidth on temperature is quite strong below 5°C but decreases rapidly with temperature and is actually quite weak for a water temperature of 20°C . So we can also obtain temperature via linewidth measurements [19, 24]. Table 1 shows the Brillouin shift ν_B and linewidth Γ_B measured by the traditional methods. The temperatures T_V and T_I in the table are respectively retrieved by the Brillouin shift and the linewidth, and the measurement accuracy can

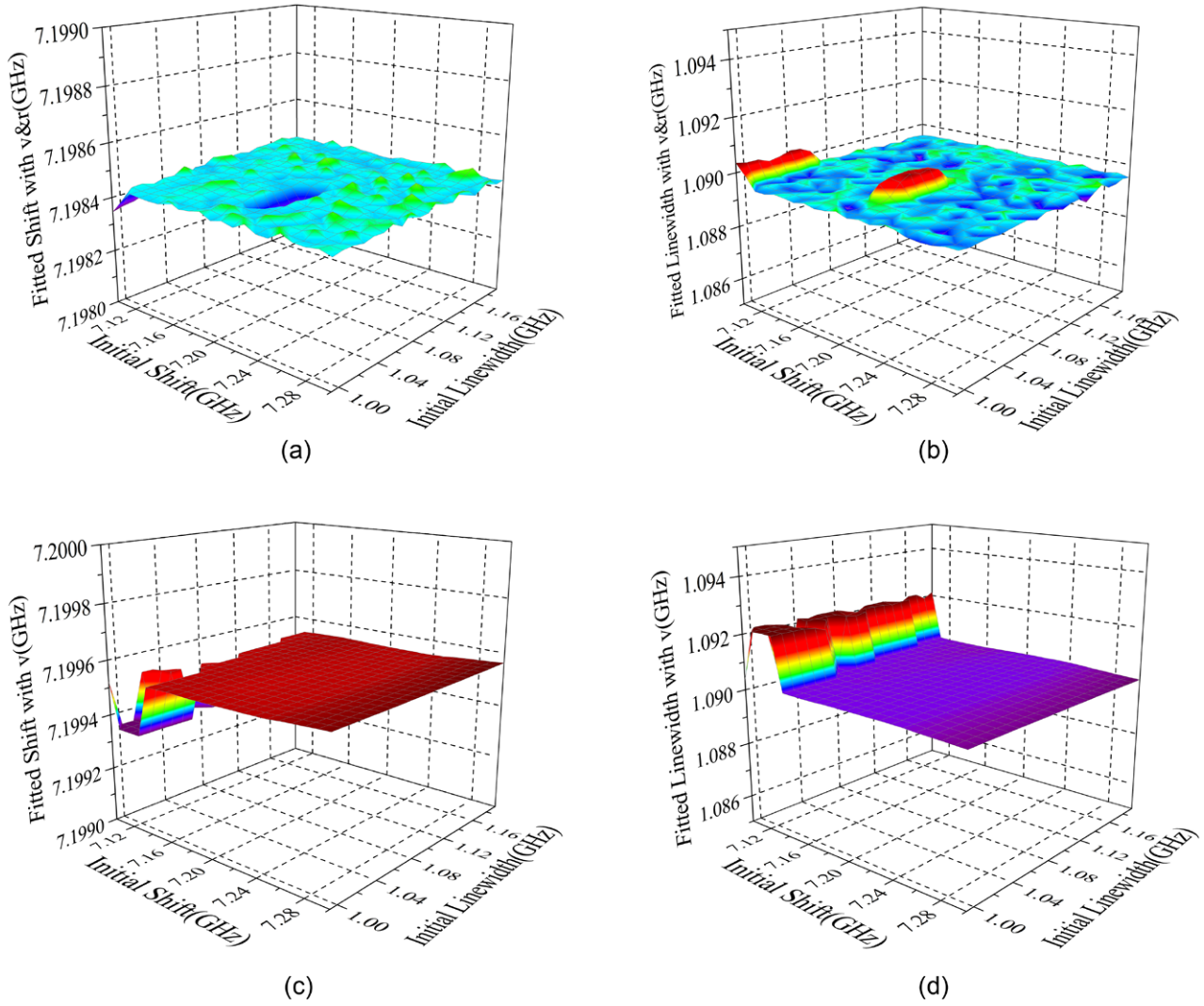


Figure 5. (a) The measured Brillouin shifts in the test initial region by fitting the spectrum on radius; (b) the measured Brillouin linewidths in the test initial region by fitting the spectrum on radius; (c) the measured Brillouin shifts in the test initial region by fitting the spectrum on frequency; (d) the measured Brillouin linewidths in the test initial region by fitting the spectrum on frequency.

be evaluated by comparing T_V with T_Γ (their differences are given in table 1 as ΔT), as the temperature calculated by ν_B or Γ_B should be of the same value in theory. The salinity S is obtained by using the dual-parameter retrieving method in [13]. Compared with true salinity 0 of the clean water used in the experiment, the difference can also be used to evaluate the measurement error.

As shown in table 1, T_V and T_Γ are the temperatures retrieved by the Brillouin shift and the linewidth, respectively. ΔT is the error between T_V and T_Γ . S is the retrieved salinity. It can be seen that the error is 3.89 °C, which corresponds to about 80 MHz error on the Brillouin shift and more than 150 MHz error on the Brillouin linewidth. For the salinity measurement, the salinity in the experiment is 0, and the measurement error is about 10.31‰. Compared with the requirement of 0.5‰ accuracy in ocean remote sensing, this measurement error is obviously too large.

By analyzing the reason for the error, it is found that three factors affect it: the limitation of the image resolution, the noise effect and the instrument broadening of the etalon.

Table 2. The measured Brillouin shift and linewidth by different fitting methods.

	Fitting on radius		Fitting on frequency	
	ν_B (GHz)	Γ_B (GHz)	ν_B (GHz)	Γ_B (GHz)
Exp 1	7.198	1.090	7.200	1.090
Exp 2	7.314	0.790	7.314	0.789
Exp 3	7.225	1.012	7.226	1.011

The resolution and noise effect have been proven to be the major effects on the accuracy. In our previous work [17], a method for improving the interference spectrum resolution was proposed. According to the symmetrical characteristics of the rings, an image processing method named ‘data folding’ combined with an ideal low-pass filter was proposed to increase the spectrum resolution and reduce the spectrum noise. By using these methods, the interference spectrum $I_{\text{rec}}(r)$ on the radius is shown in figure 4(a). This spectrum contains 11049 points in about 350 pixels, and each pixel is proven to represent 114.8 MHz frequency accuracy, so the

Table 3. Temperatures T_V and T_I respectively retrieved by Brillouin shift and linewidth, difference $\Delta T = T_V - T_I$, and salinity S with different fitting methods.

	Fitting on radius				Fitting on frequency			
	T_V (°C)	T_I (°C)	ΔT (°C)	S (‰)	T_V (°C)	T_I (°C)	ΔT (°C)	S (‰)
Exp 1	6.74	6.85	0.11	−0.404	6.83	6.85	0.02	−0.155
Exp 2	12.67	12.55	0.12	0.024	12.67	12.57	0.10	−0.036
Exp 3	8.05	8.03	0.02	−0.194	8.09	8.06	0.03	−0.111

Table 4. Detailed time costs of the fitting method on radius.

Fitting on radius	Total costs (s)	Airy(v, r) (relative)		$I(v)$ (relative)	Integration (relative)
		Sine(v, r)	Others		
Exp 1	161.07	51.03%	46.31%	0.04%	2.61%
Exp 2	120.17	48.67%	48.60%	0.05%	2.68%
Exp 3	153.04	48.46%	48.86%	0.04%	2.63%

spectrum resolution is improved to about 3 MHz per point. Also, by using the relations between the radius and the frequency in equation (4), $I_{\text{rec}}(v)$ on the frequency is obtained as figure 4(b). The spectrum in figure 4(b) only contains one pair of peaks, as the multi-order peaks are produced by the F–P interference procedure.

After the processing, the instrument broadening of the etalon still remains and affects the line shapes of both spectra. Fitting procedures respectively with equations (1) and (7) have considered the broadening effect, and are carried out for the measurement. The fitting procedures are held with the Levenberg–Marquardt (L–M) algorithm, where the Brillouin shift and linewidth are set as the fitted parameters. For the L–M algorithm, selecting suitable initial values for fitted parameters is very important, as it determines whether the fitting could reach the optimal solution. In our fitting, as the instrument broadening of the etalon mainly affects the Brillouin linewidth and rarely shows an effect on the Brillouin shift, the initial Brillouin shift is obtained by traditional methods with equation (4) on the processed spectrum. Though the value is still limited by the spectrum accuracy, it provides the value relatively close to the true values. For the initial linewidth, the value cannot be obtained by equation (4), as it will introduce a system error due to the broadening effect. To avoid this effect, the initial linewidth is replaced by the theoretical value corresponding to the temperature retrieved by the Brillouin shift [18, 19], as the temperature respectively retrieved by the Brillouin shift and linewidth should be the same. By this selection, the initial values for the Brillouin shift and linewidth are 7.200 GHz and 1.093 GHz.

Further, to evaluate whether the initial values can lead the fitting to the right optimal values, the initial Brillouin shift and linewidth are tested in the regions of 7.200 ± 0.100 GHz and 1.093 ± 0.100 GHz to evaluate the risk of multiple solutions. As the spectrum resolution is about 3 MHz, and the initial values of the Brillouin shift and linewidth will float within no more than 10 MHz, a 100 MHz region is enough to test the risk of multiple solutions. Figures 5(a) and (b) show the measured Brillouin shifts and linewidth tested in the initial region

by fitting the spectrum on the radius with equation (1) (short for ‘fitting on radius’). The measured values in the initial region by fitting the spectrum on frequency with equation (7) (short for ‘fitting on frequency’) are shown in figures 5(c) and (d). The values are tested with a 10 MHz step, as this step is enough to ensure the uniqueness of the solution for the measurement accuracy of MHz magnitude. The accuracy of the measured values in these figures is kept higher than the MHz magnitude for conveniently showing the variance (the following figures are similarly processed), while it is not necessary to keep such a lot of significant figures in the experimental result (in tables 2 and 3), as the measurement accuracy is in MHz magnitude.

In figures 5(a) and (c), the measured Brillouin shifts by fitting on radius are from 7.198 GHz to 7.199 GHz, and those by fitting on frequency are from 7.199 to 7.200 GHz, and the maximum error between both methods is less than 2 MHz. It shows that both fitting methods can obtain the stable Brillouin shifts without the risk of multiple solutions. For Brillouin linewidths in figures 5(b) and (d), the values obtained by fitting on radius are from 1.090 to 1.091 GHz; those obtained by fitting on frequency are a little larger, from 1.090 to 1.092 GHz, and the maximum error is about 2 MHz. It also indicates that both fitting methods can obtain the stable Brillouin linewidth, and no multiple solutions exist. Therefore, it can be concluded from the above analysis that both fitting procedures have good robustness with various initial values, and both measure the Brillouin shift and linewidth stably.

For the initial values of 7.200 GHz and 1.093 GHz, the measured Brillouin shift and linewidth by both fittings are used to retrieve the temperature and salinity [13, 18, 19]. The measured Brillouin shift and linewidth are listed in table 2, and the retrieved temperature and salinity are listed in table 3. Further, two more groups of experimental data are also measured to evaluate the robustness of both fitting methods, and the measured and retrieved values are also listed in tables 2 and 3, respectively.

In table 2, the maximum difference between the measured values is only 2 MHz on the measurement of the Brillouin shift in experiment 1, which indicates that two fitting methods have almost the same accuracy. Further, by comparing the temperatures and salinities in table 3, it can be found that the difference ΔT between the T_V and T_I are all less than or around 0.1 °C. Compared with the MHz magnitude or 0.5 °C requirement, both methods are sufficient for the applications. For the salinity measurement, as the experiments are held in clean water, the true value of the salinity is 0, and the values in the table are also the measurement errors. It can also be

Table 5. Detailed time costs of the fitting method on frequency.

Fitting on frequency	Total costs (s)	$r \rightarrow \nu$ (s)	Airy(ν) (relative)		$I(\nu)$ (relative)	Convolution (relative)
			Sine(ν)	Others		
Exp 1	2.81	0.97	1.42%	0.71%	0.71%	62.63%
Exp 2	2.86	0.93	1.40%	1.40%	0.70%	63.99%
Exp 3	2.73	0.92	1.47%	1.10%	0.73%	63.00%

found that the errors are all below 0.5‰, which is proven to be the variance of the salinity distribution of the ocean water [18]. Therefore, for ocean salinity remote sensing, both methods can also work well. From the above analysis on the measurement accuracy, it can be concluded that both methods can obtain the sufficiently accurate temperature and salinity of the ocean.

We also evaluated the time costs of both fitting methods. This evaluation is performed on a computer with a dual-core central processing unit (each core has a frequency of 3 GHz) and a memory of 4 GB. The results are shown in tables 4 and 5.

The $r \rightarrow \nu$ in table 5 means the time cost on transforming $I_{\text{rec}}(r)$ to $I_{\text{rec}}(\nu)$. In table 4, it can be seen that the total time costs are all over 100 s, while the largest proportion of time cost is the calculation of the sine function and other parts of Airy(ν, r), which is more than 97% of the total computation time. It proves that the rapid measurement capability of the fitting on radius is mainly limited by calculating Airy(ν, r), and the reason is that the double variables exponentially increase the calculating times of Airy(ν, r). For the fitting on frequency, it can be found in table 5 that the total time cost is reduced to less than 3 s. This is because the Airy function is only based on frequency after transforming $I_{\text{rec}}(r)$ to $I_{\text{rec}}(\nu)$, and the calculating times are greatly reduced, so the time cost of the Airy function is reduced to the same level with $I(\nu)$, while the cost just adds a transforming procedure of less than 1 s. Compared with the more than 100 s fitting on radius, the fitting on frequency is much faster.

Above analysis shows that the fitting methods based on equation (1) and based on equation (7) can both reach the temperature measurement accuracy of 0.1 °C and salinity measurement accuracy of 0.5‰. However the latter is more suitable for the rapid measurement application because it is less time consuming.

5. Conclusion

When the Brillouin lidar system with the F-P etalon is applied in remote sensing the ocean physical characteristics, the transmission function of the F-P etalon will affect the measurement accuracy of the Brillouin scattering spectrum characteristics by changing the line shape of the spectrum. Two fitting methods, the integration method on radius and the convolution method on frequency, have been proposed to remove the effect of the transmission function and ensure the measurement accuracy. Analysis for the measurement accuracy and the time costs of the two fitting methods has been discussed and compared. Theoretical analysis and experimental results showed that both fitting methods can effectively reduce the

error due to the transmission function of the F-P etalon and ensure the MHz measurement accuracy on spectrum characteristics, 0.1 °C on temperature and 0.5‰ on salinity, while the convolution method on frequency has more capability on rapid measurement, which is sufficient for the on-line remote sensing.

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